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Dietary supplementation of clinically utilized PI3K p110α inhibitor extends the lifespan of male and female mice

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Diminished insulin and insulin-like growth factor-1 signaling extends the lifespan of invertebrates[1–4]; however, whether it is a feasible longevity target in mammals is less clear[5–12]. Clinically utilized therapeutics that target this pathway, such as small-molecule inhibitors of phosphoinositide 3-kinase p110α (PI3Ki), provide a translatable approach to studying the impact of these pathways on aging. Here, we provide evidence that dietary supplementation with the PI3Ki alpelisib from middle age extends the median and maximal lifespan of mice, an effect that was more pronounced in females. While long-term PI3Ki treatment was well tolerated and led to greater strength and balance, negative impacts on common human aging markers, including reductions in bone mass and mild hyperglycemia, were also evident. These results suggest that while pharmacological suppression of insulin receptor (IR)/insulin-like growth factor receptor (IGFR) targets could represent a promising approach to delaying some aspects of aging, caution should be taken in translation to humans.

Suppression of the IR/IGFR signaling pathway consistently extends the lifespan of *Caenorhabditis elegans* and *Drosophila melanogaster*[1–4]; however, results from mouse models have been less consistent. Complete ablation of the IR[5] or IGF1 receptor globally[6–8] results in premature death, and while female IGF1 receptor haploinsufficient mice were originally reported to live 33% longer than controls[7], subsequent studies reported lesser effects and even reduced lifespan[9–12]. Adult-induced partial peripheral tissue IR haploinsufficiency did not affect lifespan[8]. While adipocyte-targeted deletion of the IR appeared to extend lifespan[13], other adipose IR deletion studies resulted in lipodystrophy and severe liver disease, reducing lifespan[14,15]. Similar inconsistent results are seen in models of suppressed IR substrate 1 and 2 signaling[16–18].

IR and IGFR activation initiates phosphorylation events that activate phosphoinositide 3-kinase (PI3K), which then coordinates nutrient metabolism and growth[19]. Extended lifespans and healthspans are seen following targeted genetic suppression of PI3K in middle-aged male mice heterozygous for the kinase-dead PI3K p110α[20], in overexpression of PTEN (a phosphatase that counteracts the activity of the class

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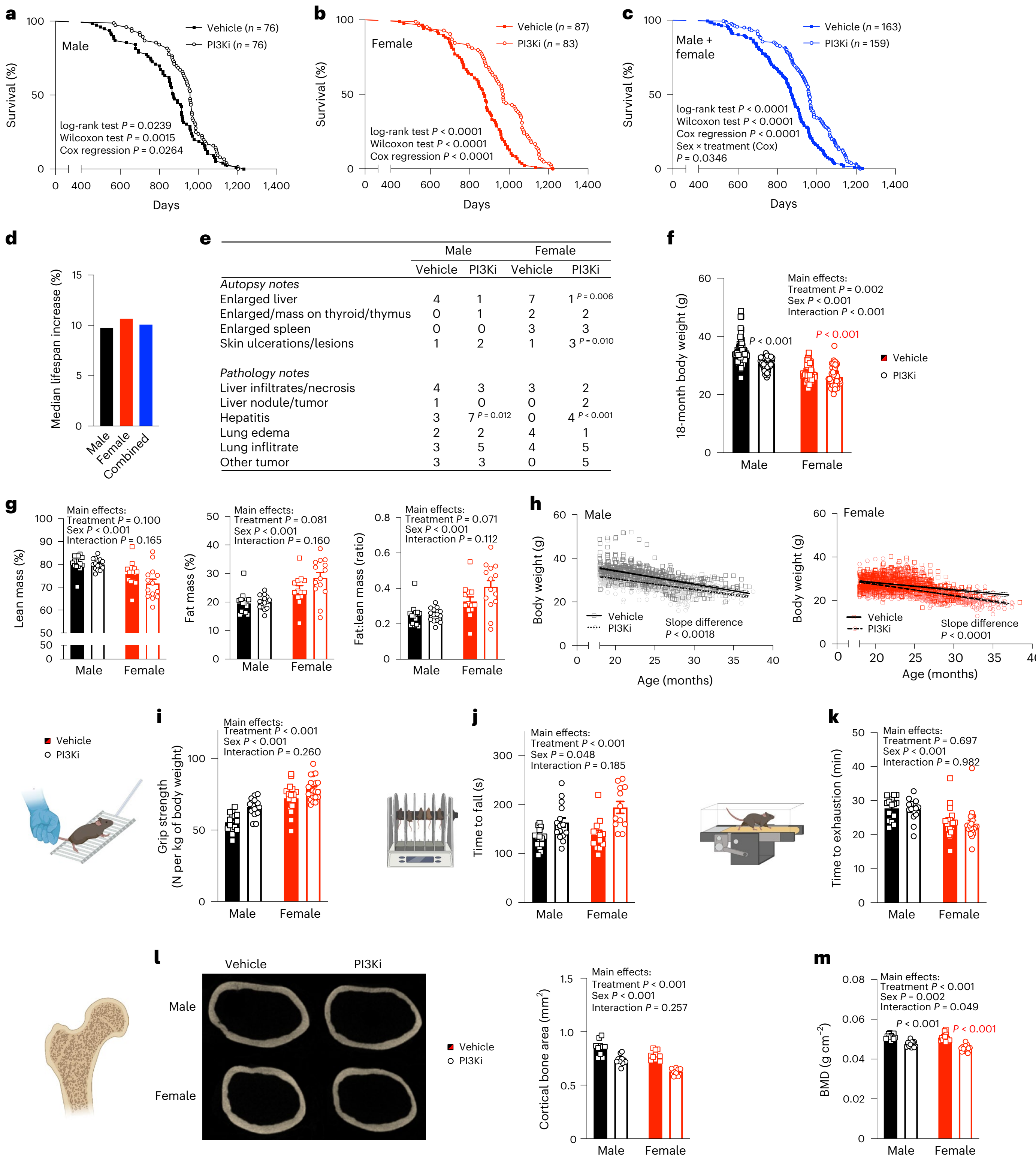

	Male		Female	
	Vehicle	PI3Ki	Vehicle	PI3Ki
Autopsy notes				
Enlarged liver	4	1	7	1 $^{P=0.006}$
Enlarged/mass on thyroid/thymus	0	1	2	2
Enlarged spleen	0	0	3	3
Skin ulcerations/lesions	1	2	1	3 $^{P=0.010}$
Pathology notes				
Liver infiltrates/necrosis	4	3	3	2
Liver nodule/tumor	1	0	0	2
Hepatitis	3	7 $^{P=0.012}$	0	4 $^{P<0.001}$
Lung edema	2	2	4	1
Lung inflitrate	3	5	4	5
Other tumor	3	3	0	5

Fig. 1 | Diet supplemented with a PI3Ki extends the lifespan of mice. **a**–**c**, Lifespan survival curves for male (**a**), female (**b**), and combined male & female (**c**) mice. **d**, Effect of PI3Ki on median lifespan. **e**, Morbidity and autopsy observations ($n = 20$ per group/sex). **f**,**g**, Body mass (18 months) and composition (20 months). **h**, Body mass from 18 months. **i**–**k**, Grip strength, rotarod and treadmill running performance at 24 months of age. **l**, Cross-section and cortical area of the femur. **m**, total body bone mineral density by dual X-ray absorbance (DXA) scan at 20 months. Results are mean ± s.e.m., with biological replicates (individual mice, n) shown as individual data points or stated within the figure. Significance was determined by log-rank test, Wilcoxon test and Cox regression analyses (**a**–**d**); chi-squared test (**e**); two-way ANOVA with Sidak post hoc analysis for within-sex effects when a significant sex × treatment interaction was seen (**f**,**g**,**i**–**m**) or linear regression (**h**). The P values within figures are for PI3Ki versus vehicle of the same sex. BMD, bone mineral density.

I PI3Ks) in both male and female mice, and in mice with Akt1 (a PI3K target) haploinsufficiency[21,22]. The development of therapeutic small-molecule PI3K inhibitors has provided the opportunity to determine whether pharmacological approaches to suppress PI3K signaling can also promote healthy aging. Alpelisib (BYL719, known commercially as Piqray)[23] is an orally available PI3K p110α-selective inhibitor that has been approved by the US Food and Drug Administration for treating HR+/HER2− advanced breast cancer with a PIK3CA mutation and has shown some clinical efficacy in the treatment of PIK3CA-related overgrowth syndromes[24]. Here, we use alpelisib to investigate the hypothesis that chronic pharmacological inhibition of PI3Kα would extend the healthspan and lifespan of mice.

Previously, we have reported the pharmacokinetic profile of a diet-delivered PI3Ki[25] and showed that in middle-aged (8-month-old) mice, a dose of 0.3-mg kg^{-1} diet leads to a plasma concentration of 5 μM, moderate hyperinsulinemia, hyperglycemia and glucose intolerance for at least 6 weeks[25,26]. Based on these observations that diet-delivered PI3Ki leads to sustained on-target effects (impaired insulin sensitivity), we chronically exposed male and female C57BL/6J mice to a chow diet containing a vehicle or PI3Ki from 12 months of age. At 20 months of age, alpelisib concentrations were 0.7–6.5 μM in plasma and 4.5–29.5 μmol kg^{-1} in liver of PI3Ki-treated mice and undetected in vehicle mice (Extended Data Fig. 1a). PI3Ki-treated mice had a median lifespan that was 10% longer than vehicle-treated mice (Fig. 1a–d). Log-rank test, Wilcoxon test and Cox regression analyses indicated significant differences between vehicle and PI3Ki for both sexes independently, but the lifespan extension was more pronounced in females (sex by treatment interaction by Cox regression P = 0.035). Maximal lifespan (90th percentile, Boschloo's test) was extended by PI3Ki in the entire cohort (n = 6 vehicle and n = 27 PI3Ki mice in the 90th percentile; P = 0.001) and independently in females (n = 2 vehicle and n = 15 PI3Ki mice in the 90th percentile; P = 0.002) but not males (n = 4 vehicle and n = 11 PI3Ki mice in the 90th percentile; P = 0.068). PI3Ki did not affect the morbidity rate or have a consistent impact on male and female gross autopsy or pathology observation at death (Fig. 1e), although male and female PI3Ki-treated mice had higher rates of what appears to be hepatitis. This suggest that PI3Ki did not prevent a specific pathology or cause of death common to the organs assessed.

Independent of food intake, PI3Ki led to a loss of body weight that was evident by one month of treatment in males and six months in females (Fig. 1f and Extended Data Fig. 1b,c). Vehicle and PI3Ki mice had similar relative (to total mass) lean and fat mass and fat-to-lean mass ratio (Fig. 1g). Mouse body weights tend to decrease from ~20 months of age[27], and this age-associated loss of body mass/sarcopenia is associated with increased risk of death and reduced healthspan[27,28]. Therefore, we next assessed whether PI3Ki altered the rate of weight loss from 20 months of age. While male PI3Ki mice had a slower age-related loss of body weight than vehicle mice, female PI3Ki mice had an increased rate, and PI3Ki treatment did not affect the rate of weight loss in the last four months of life in either sex (Fig. 1h and Extended Data Fig. 1d). This raises the question of whether PI3Ki was extending healthspan as well as lifespan. To address this, physical, metabolic and cognitive tests were undertaken in mice aged 18–24 months. PI3Ki-treated mice of both sexes showed signs of improved healthspan with increased grip strength and balance and/or coordination (rotarod test) at 24 months (Fig. 1i,j). However, PI3Ki treatment did not affect exercise capacity (treadmill running to exhaustion) (Fig. 1k) or measures of cognitive function (novel object and open field tests) (Extended Data Fig. 1e,f). In males, PI3Ki treatment did not affect any echocardiography indices of cardiac function (Extended Data Fig. 1g). Consistent with previous observations in young rats[29], PI3Ki mice had reduced total body bone mineral density and reduced cortical bone size, area and thickness, indicating the promotion of osteoporosis (Fig. 1l,m).

Improved insulin sensitivity is seen as a marker of healthy aging[30], and interventions that improve insulin sensitivity[31–33], including some that inhibit PI3K signaling[20,21], extend lifespan of mice. Therefore, we assessed the impact of PI3Ki treatment on whole-body glucose homeostasis and tissue-specific insulin signaling. At 18 months, PI3Ki mice had elevated plasma insulin, IGF1 and blood glucose (Fig. 2a,b and Extended Data Fig. 2a). Blood glucose response to a glucose (at 24 and 30 months) or insulin (at 24 months) bolus was impaired by PI3Ki treatment (Fig. 2c,d and Extended Data Fig. 2b). Indicative of a sexual dimorphic glucose homeostasis response, PI3Ki-induced glucose intolerance was greater in females than in males (interaction P < 0.01); however, both sexes showed tissue-specific impaired insulin signaling as evidenced by elevated insulin receptor phosphorylation in the absence of concurrent increases in Akt or FOXO1 phosphorylation (Fig. 2e and Extended Data Figs. 3 and 4a,b). Despite this, PI3Ki-treated mice were able to normally regulate their blood glucose response to a physiological meal challenge and did not have chronically pathologically elevated blood glucose, as indicated by similar plasma fructosamine (a marker of glycated protein formation) (Extended Data Fig. 2c,d) between groups. While phosphorylation of p70S6, a downstream target of nutrient-sensing kinase mTOR, was lower in livers from male and female PI3Ki mice, this effect was not seen in muscle or white or brown adipose tissue of either sex (Fig. 2e and Extended Data Fig. 3).

While some sex-related differences in PI3Ki-associated aging phenotypes were observed, generally the direction of change seen in significantly affected phenotypes was similar for males and females (Fig. 2f), suggestive of a common underlying pathway. Lowering oxidative stress and cellular senescence/inflammation are common mechanisms through which suppressing insulin/IGF1 signaling is thought to prolong lifespan[7,11,34–36]. However, this mechanism was not evident as vehicle and PI3Ki-treated mice had similar plasma inflammatory cytokine markers (Fig. 2g) and markers of oxidative stress (plasma protein carbonyls, hepatic nuclear and mitochondrial DNA damage) (Fig. 2h–j). Similarly, PI3Ki and vehicle male and female mice had similar hepatic ULK1 phosphorylation and expression of p62 and LC3 II/I ratio in liver and white adipose tissue (WAT), suggesting that PI3Ki treatment was not having a large impact on autophagy (Extended Data Fig. 4c,d). mRNA microarray of livers revealed little regulation of genes associated with oxidative stress or senescence by PI3Ki (Extended Data Fig. 5). Overall, there was a very limited impact of PI3Ki on the hepatic transcriptome, with only 49 of 21,958 genes being differentially regulated in either male or female mice. Of these, only three, *Egfr*, *Esm1* and *Lepr*, were affected (all increased more than twofold, P < 0.05) in both males and females (Fig. 2k).

Alpelisib and dual PI3Kα/PI3Kδ inhibitors induce fat loss in obese mice[37,38], and adipose-specific p110α knockout mice gain less weight with aging[39]. Since PI3Ki-induced weight loss was evident in both sexes, we turned our attention to adipose tissue. PI3Ki-treated mice had larger subscapular brown adipose tissue (BAT) and smaller subcutaneous white adipose tissue (sWAT) fat pads (Fig. 3a). In males, PI3Ki-treated mice had smaller sWAT adipocyte size, and both sexes had elevated plasma leptin (Fig. 3b,c and Extended Data Fig. 6a–c). Microarray mRNA analysis of male sWAT revealed an enrichment of genes involved in mitochondrial aerobic metabolism (Extended Data Fig. 6d,e), and 25% of all genes with significantly altered mRNA expression localized to the mitochondria (as annotated in MitoCarta3[40]) (Fig. 3d). Consistent with evidence that reducing IR expression in adipose tissue can increase mitochondrial volume[8,41], sWAT from aged male PI3Ki-treated mice had increased capacity to utilize oxygen during both oxidative phosphorylation and uncoupled respiration (Fig. 3e). While interventions that improve mitochondrial function are thought to promote healthy aging[42] and contribute to the metabolic benefits of suppressing IR/IGF1R signaling[35,41,43], it may not be the underlying mechanism here since sWAT from PI3Ki-treated female mice did not show any enrichment of genes in mitochondrial or energy pathways (Extended Data Fig. 6f,g), and in middle-aged mice, PI3Ki does not impact mitochondrial function in nonadipose tissue (muscle or liver)[26].

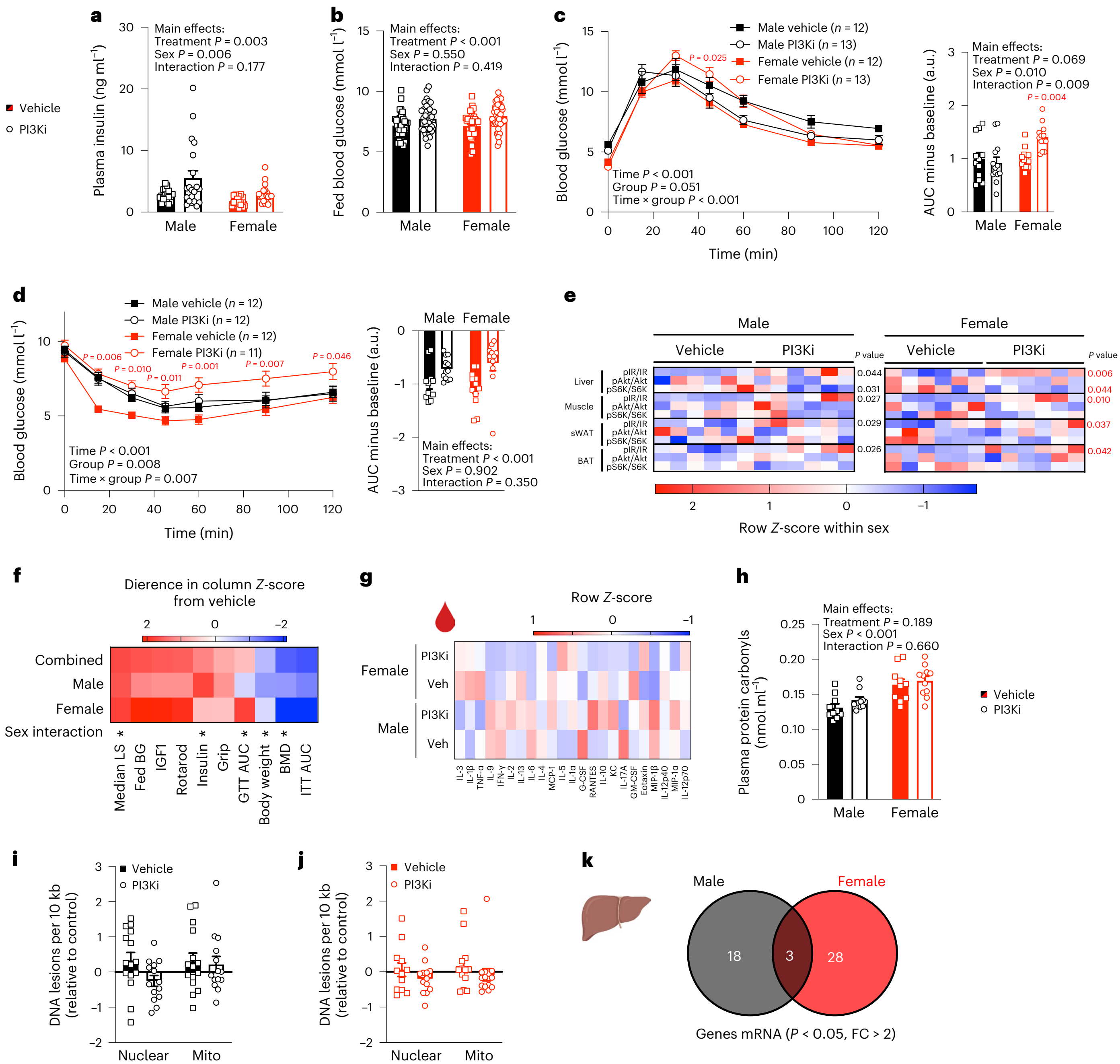

Fig. 2 | Effect of PI3Ki on glucose homeostasis, oxidative stress and inflammation with aging. a,**b**, Plasma insulin (**a**) and blood glucose (**b**) at 18 months. **c**,**d**, Glucose (**c**) and insulin (**d**) tolerance tests at 20 months. **e**, Basal protein expression and phosphorylation of insulin signaling intermediates in tissues collected at 20 months ($n = 6$ per group except for liver pAkt, which is representative for $n = 12$ per group). Refer to Extended Data Fig. 3 for the original immunoblot data used to create the heat map. **f**–**g**, Heat maps of group mean *Z*-scores for phenotypes significantly affected by PI3Ki treatment (**f**) and plasma cytokines (**g**) measured at 20 months ($n = 11$ per group). **h**–**j**, Plasma protein carbonyls (**h**) and liver nuclear and mitochondrial DNA damage in male (**i**) and female (**j**) mice at 20 months. **k**, Number of genes in PI3Ki-treated liver samples with expression that was significantly different from vehicle controls as assessed by mRNA microarray ($n = 3$ per group pooled from two to three independent mice). Results are mean ± s.e.m. with biological replicates (individual mice, *n*) shown as individual data points or stated within the figure. Significance was determined by two-way ANOVA (**a**–**h**) or RM two-way ANOVA (**c**,**d**) with Sidak post hoc analysis for within-sex effects when a significant sex × treatment interaction was seen or an unpaired two-sided Student's *t* test (**i**,**j**). *P* values within figures are for PI3Ki versus vehicle of the same sex. RM, repeated measures; a.u., arbitrary unit; AUC, area under the curve; BG, blood glucose; FC, fold change; GTT, glucose tolerance test; ITT, insulin tolerance test; LS, lifespan; *, $P < 0.05$.

There was a clear sexual dimorphic impact of PI3Ki on sWAT gene expression, with only 8 of 200 significantly altered genes being observed in both males and females, and no overlapping gene ontology biological processes were evident (Extended Data Fig. 6h). In contrast, 17% of BAT genes in which expression was altered by PI3Ki were common in males and females, and these were enriched for metabolic processes associated with lipid and ketone metabolism (Fig. 3f–h and Extended Data Fig. 7a,b). Consistent with potentially altered BAT substrate metabolism, PI3Ki-treated mice had lower plasma triglycerides and increased overnight fasted plasma FFA (Fig. 3i and Extended Data

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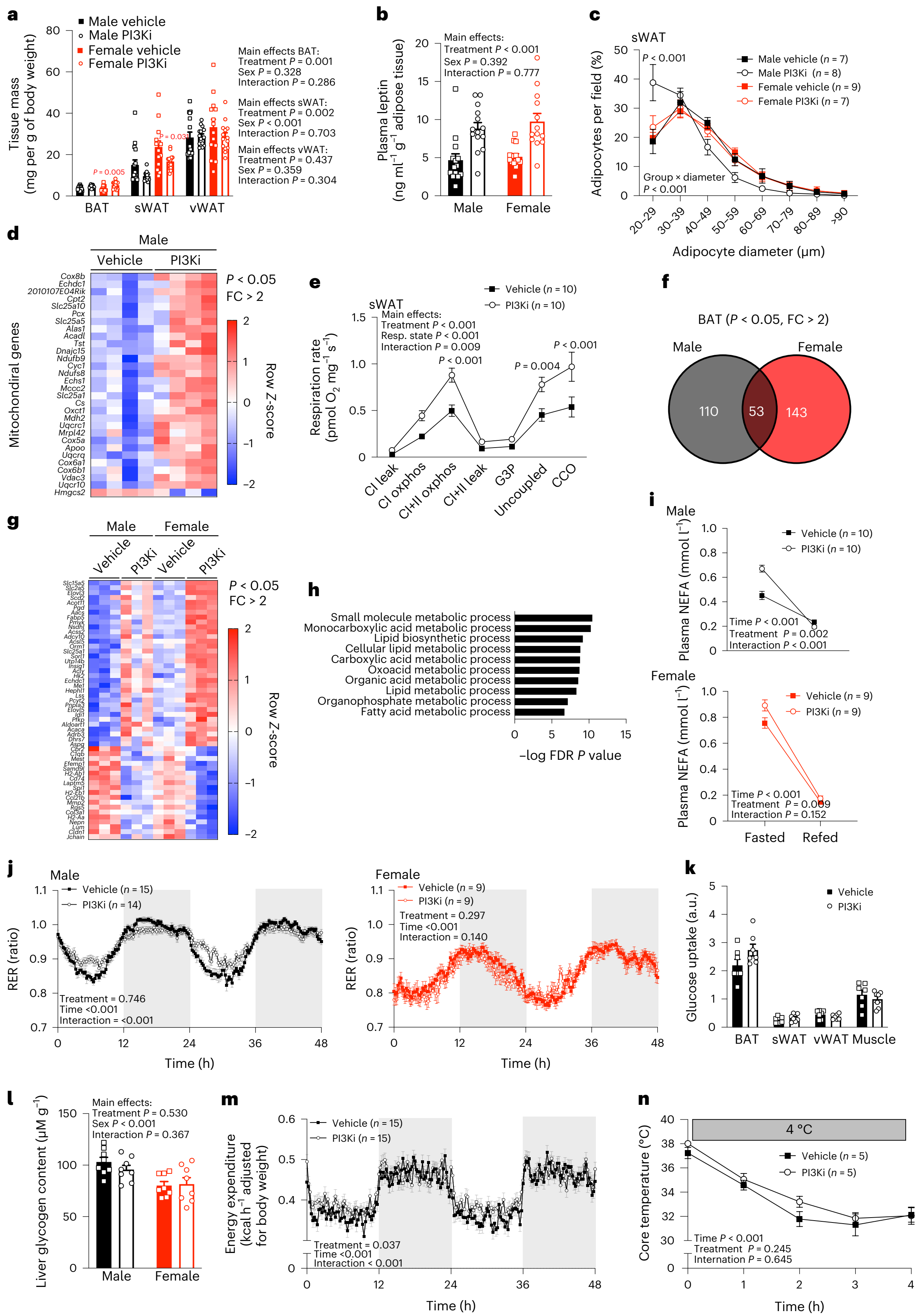

Fig. 3 | PI3Ki increases BAT lipid metabolism pathways and improves WAT mitochondrial function in males during aging. **a**, Adipose tissue mass, **b**,**c**, plasma leptin per gram of WAT (**b**) and sWAT adipocyte diameter (**c**) at 20 months. **d**, Twenty-month-old male PI3Ki sWAT mitochondrial-related genes with expression that was significantly different from vehicle (by mRNA microarray, $n = 4$ per group pooled from two to three independent mice). **e**, Mitochondrial oxygen consumption capacity. **f**, Overlap between male and female and heat map (**g**) of genes in BAT from PI3Ki with expression that was significantly different (by mRNA microarray, $n = 3$ per group pooled from two to three independent mice). **h**, The top ten male and female overlapping gene ontology biological processes upregulated in PI3Ki. **i**, Overnight fasted and 1-h refed plasma free fatty acids from 20-month-old male and 6-month-old female mice. **j**, Respiratory exchange ratio (RER) from 20-month-old male and 6-month-old female mice. **k**–**n**, Tissue glucose uptake (**k**), liver glycogen (**l**), energy expenditure (**m**) and the core body temperature during cooling (4 °C) (**n**) of male mice at 20 months. Results are mean ± s.e.m. with biological replicates (individual mice, n) shown as individual data points or stated within the figure. Significance was determined by two-way ANOVA (**a**–**c**,**e**,**i**,**k**,**l**), RM two-way ANOVA (**n**) or mixed linear model (**j**,**m**). The P values within the figures are for PI3Ki versus vehicle of the same sex. RM, repeated measures; a.u., arbitrary unit; BAT, brown adipose tissue; sWAT, subcutaneous white adipose tissue; vWAT, visceral white adipose tissue; FDR, false discovery rate; CI, complex I; CII, complex II; G3P, glycerol-3-phosphate; CCO, cytochrome-c oxidase; FC, fold change.

Fig. 7c). In 20-month-old males but not 6-month-old females, PI3Ki treatment reduced circadian oscillations in whole-body substrate metabolism (RER) (Fig. 3j). These effects in males did not appear to be the direct effect of restricted availability of glucose for utilization because vehicle and PI3Ki-treated mice had similar WAT, BAT and muscle glucose transport, liver glycogen stores (Fig. 3k,l), glucose de novo lipogenesis in WAT and hepatic phosphorylation of the energy-sensing kinase AMPK (Extended Data Figs. 4b and 7d). Nor did the impact of PI3Ki treatment have a substantial impact on energy expenditure, with only a small increase during the light phase in males and a similar reduction in core temperature during cooling (6 h at 4 °C) being evident in vehicle and PI3Ki (Fig. 3m,n and Extended Data Fig. 7e). This suggests that, unlike during obesity or acute PI3Ki treatment[21,37], chronic PI3Ki inhibitor treatment during aging did not increase thermogenic energy loss. This is consistent with BAT mitochondrial oxidative capacity being similar in the vehicle and PI3K-treated male mice (Extended Data Fig. 7f).

Discussion

Here, we provide evidence that treatment with a clinically utilized small-molecule inhibitor of PI3K p110α (alpelisib) from middle age can extend the median and maximal lifespan of mice. Only a few pharmacological treatments have been shown to consistently extend the lifespan of both male and female mice[44–46], with the mTOR inhibitor rapamycin being the lead compound[47,48]. Growth factors can act via PI3K to activate mTOR[19], and aged PI3Ki-treated mice showed a phenotype similar to rapamycin-treated mice in terms of reduced body mass, insulin resistance and greater lifespan extension in females[47,48]. Furthermore, PI3Ki lowered activation (phosphorylation) of the mTOR downstream target p70 S6K in the liver. While this may suggest that the lifespan-extending effect of PI3Ki treatment could be attributed to mTOR inhibition, this seems unlikely given that mTOR inhibition was not evident in other tissues (fat and muscle), and PI3Ki had little impact on the hepatic transcriptional profile or markers of oxidative stress or inflammation, which are thought to at least partially explain the age-delaying benefits of rapamycin/rapalogs in mice[48–50].

Calorie restriction and associated fasting are the most effective interventions to extend the lifespan of mammals[51,52]. It is tempting to speculate that PI3Ki may be reducing glucose utilization and thus, acting as a calorie restriction mimetic; however, our data do not support this either. Despite fed blood glucose being mildly elevated by PI3Ki treatment and females being intolerant to a glucose bolus, when treated with a PI3Ki mice showed normal blood glucose control in response to a physiological (meal) glucose challenge and had similar liver glycogen stores and tissue glucose uptake, suggesting overall a limited impairment in glucose metabolism. However, PI3Ki-treated mice had differential circadian oscillations in substrate metabolism and enhanced fasting-induced lipolysis. Given that PI3Ki upregulated BAT transcription pathways relating to lipid metabolism in both sexes and that there is building evidence of age-associated alteration in BAT lipid species and rhythmicity of lipid metabolism[53,54], altered BAT metabolism may contribute to the longevity phenotype observed in PI3Ki-treated mice. Indeed, improved BAT function or volume has been reported in other long-lived murine models[21,55–58], including mice that overexpress *Pten*, a negative regulator of PI3K signaling[21]. If altered BAT function does underpin the beneficial effect of PI3Ki on aging, this may be occurring independent of its thermogenic role, with PI3Ki showing similar energy expenditure and cold-induced thermogenesis as vehicle-treated mice.

While PI3Ki treatment improved some markers of healthspan, including strength and balance and coordination, it also had no impact or even negative impacts on other markers of aging like hyperglycemia, cognition, cardiovascular function and bone mass. This is important to highlight when considering the translation of these findings as dementia and osteoporosis are major age-related illnesses in humans that reduce healthspan and lifespan. Recently, elevated blood glucose has been shown to associate with lower mortality risk in mice, while the relationship is reversed in humans and primates[27]. While consistent with the observation that female mice had greater PI3Ki-mediated glucose intolerance and longevity, it also questions the translatability of interventions that elevate blood glucose to delay aging in humans. In any case, if the hyperglycemic on-target effect of insulin-signaling intermediate inhibition was detrimental, it could be controlled using SGLT2 inhibitors, which act to lower blood glucose by increasing urinary excretion of glucose and have been shown to extend the lifespan of male mice[32].

A limitation of this study is that this was a single lifespan experiment using one strain of inbred (C57BL/6J) mice. It will be important to determine the reproducibility of the lifespan-extending effects of PI3Ki in genetically heterogenous mice housed in different facilities and the impact of different doses and types of PI3Ki. We were only able to assess cardiovascular function in male mice at 15 months of age. This is a limitation because cardiac aging in mice is often not evident until 18 months of age, and recent evidence suggests that suppression of the cardiac IGF1 pathway may have a biphasic impact on cardiovascular function, being most beneficial in 20-month-old mice[59].

In summary, this work provides evidence that long-term pharmacological targeting of PI3K p110α with alpelisib during aging is well tolerated and can extend the median lifespan of male and female mice. As alpelisib is already used to treat certain cancers and overgrowth syndrome, explorations of long-term use on biomarkers of aging in humans are warranted.

Methods

Animals

C57BL/6J mice were bred and housed in groups of 8–13 at either AgResearch Small Animal Colony (Ruakura, New Zealand) or the Vernon Jansen Unit University of Auckland (Auckland, New Zealand) in standard open top cages with enrichment (mouse house and/or tunnel), with an ambient temperature of 22–23 °C and 12-h light/dark cycles. All mice had ad libitum access to food and tap water. At 12 months of age, cages were block randomized based on body weight to PI3Ki or vehicle condition. The PI3Ki groups received a standard chow diet (Meat Free Rat and Mouse Diet; Specialty Feeds Pty, Glen Forrest, Western Australia, Australia) that contained 0.3 g per kg of the PI3Kα selective inhibitor alpelisib (also known as BYL719 and commercially as Piqray[23]; Medchem Express). Vehicle groups received the same diet without alpelisib incorporated.

All experiments described were approved by the University of Auckland Animal Ethics Committee (R001960 and R001479). Overall, experimental design is illustrated in Extended Data Fig. 8. Source data for all figures and extended data are provided with this manuscript.

Lifespan and healthspan

Animals were observed daily, and body weight was monitored monthly until either natural death or when the animal facility veterinarian deemed mice to be moribund, at which point they were euthanized and the age was recorded as lifespan. At the time points indicated, subsets of mice from each group underwent healthspan testing, and some were euthanized for tissue collection. In total, 233 male and 237 female mice entered this study; $n = 14$ or 15 per group were euthanized at ~20 months of age for tissue collection, following in vivo assessment of glucose homeostasis and cognitive/behavioral phenotype at ~18 months of age. A further $n = 10$ or 11 per group were euthanized for end point tissue analysis of adipose tissue and bone phenotype after body composition assessment at ~20 months of age. Thirty males ($n = 15$ each for vehicle and PI3Ki) were used for echocardiography and indirect calorimetry, followed by in vivo glucose uptake end point experiment at ~15 months of age. The remaining 152 males ($n = 76$ each for vehicle and PI3Ki) and 170 females ($n = 87$ vehicle and $n = 83$ PI3Ki) mice contributed to the lifespan assay. Some of these mice (n indicated in figures) undertook in vivo phenotyping of glucose homeostasis and physical and behavioral assessment at ~24 and/or 30 months of age, but they remained in the lifespan study. Eighteen female mice ($n = 9$ each for vehicle and PI3Ki) were used to perform indirect calorimetry and plasma free fatty acid analysis at six months of age, following one month of intervention treatment.

Pathology

When mice were found dead or euthanized due to morbidity, they were stored at −20 °C. Autopsy and tissue histology were performed on 20 randomly selected mice per treatment group per sex. Gross autopsy and qualitative examination of paraffin-embedded, H&E-stained sections (5 µm) of brain, heart, lung, liver, spleen, thymus, thyroid, pancreas, kidney, adrenal, gut, gonad and skin tissues were undertaken by two clinical pathologists using a multihead microscope.

Food intake and glucose homeostasis

After three months of diet intervention, average daily food intake was determined for each cage over one week. Blood was sampled from the tail vein to measure blood glucose (Accu-Chek Performa), and plasma was separated to measure insulin concentration by AlphaLISA (PerkinElmer). Whole-body glucose tolerance and insulin sensitivity were assessed by oral glucose bolus (2 mg kg^{-1}) or intraperitoneal (i.p.) insulin injection (0.7–0.8 IU kg^{-1}) following a short-term fast (4–6 h). A meal challenge was performed by fasting mice overnight, allowing access to food for 3 h, and then, refasting for 4 h. For all tests, blood glucose was monitored from the tail vein (Accu-Chek Performa) at the indicated time points.

Indirect calorimetry

Indirect calorimetry to assess energy expenditure and respiratory exchange ratio was undertaken in individually housed mice using a Promethion Core metabolic cage system (Sable Systems). Mice were acclimatized for 48 h before data were collected for 48 h. Mice had ad libitum access to tap water and their respective study diet throughout both acclimatization and data recording phases. All data were recorded continuously from 12 cages simultaneously and processed into 15-min averages using MacroInterpreter v.2.38.

Echocardiography

Animals were anaesthetized with isoflurane and transthoracic echocardiography was performed using the VEVO LAZR-X 3100 with an MX400 probe (20- to 46-Hz) linear array transducer coupled with a digital ultrasound system (FUJIFLIM VisualSonics). All analyses were performed by an investigator blinded to study group allocation. Left ventricular (LV) m-mode two-dimensional echocardiography was performed in a parasternal short-axis view at the midpapillary level to measure LV wall and chamber dimensions to derive the following parameters: percentage of ejection fraction (((end diastolic volume − end systolic volume)/end diastolic volume) × 100), cardiac output (stroke volume × heart rate) and corrected LV mass (1.053 × [LV internal diameter in diastole + LV posterior wall thickness in diastole + interventricular septum in diastole]^3 − LV internal diameter in diastole^3) × 0.8). At least three consecutive cardiac cycles were sampled per cineloop image and three images per animal were analyzed and averaged using Vevo LAB software (VisualSonics).

DXA

Mice were anesthetized by i.p. injection of sodium pentobarbitone, and body composition was determined via DXA scan (Lunar PIXImus II; GE Healthcare). Upon completion of DXA, mice were euthanized by cervical dislocation, and adipose tissue was collected to determine mitochondrial respiration and in vitro lipolysis and lipogenesis. Femurs were also collected for microcomputed tomography (micro-CT) scanning.

Femur bone micro-CT

Soft tissue was removed from the right hindlimb, and femurs were fixed in 70% ethanol at 4 °C. The distal end of each femur was scanned using a Skyscan 1172 micro-CT scanner (Bruker) as previously described[60]. The X-ray voltage was 59 kV, the amperage was 167 µA, and a 0.5-mm aluminum filter was used. Images were acquired with an isotropic voxel size of 7 µm, with 180° of rotation and a rotation step of 0.3°. After standardized reconstruction using NRecon software (Bruker, v.1.6.9.18), the datasets were analyzed using CTAn software (Bruker, v.1.18.8.0). Standardized parameters of trabecular and cortical bone microstructure were measured[61]. In females and males, the trabecular region of interest was 0.53 or 0.70 mm proximal to the growth plate, respectively, and extended 1.76 and 2.47 mm in the proximal direction, respectively. In both sexes, the cortical region of interest was 5.64 mm proximal to the growth plate and extended 0.7 mm in the proximal direction.

Cognitive and behavioral testing

General stress and anxiety behavior and locomotor activity were assessed using an open field test, and memory was assessed using a novel object recognition test essentially as described previously[25] in custom-built perspex enclosures (270 mm (length) × 265 mm (width) × 350 mm (height)) with opaque enclosure walls. For open field testing, mice were placed in the center of the enclosure, and movements were recorded for 25 min using an overhead camera (GoPro Hero 6; GoPro Inc.). Times spent in the inner and outer zones, as well as total distance moved, were analyzed using EthoVision XT 12 tracking software (Noldus Information Technology). For the novel object recognition test, two identical objects were placed in diagonally opposite corners of the enclosure 5 cm from the wall edge, and mice were placed in the enclosure and recorded for 5 min. After 4 h, mice were returned to the enclosure and recorded for 5 min with one object the same as before and one novel object, and the time spent exploring each object was determined (EthoVision XT 12 tracking software; Noldus Information Technology).

Physical performance tests

Grip strength, rotarod (motor coordination) and treadmill running capacity were undertaken as described previously[25,26]. Mice were familiarized with all physical tests at least twice a minimum of 48 h before recorded measures. For four limb grip strength assessments, mice were placed on the grid of a BioSeb Grip Strength Meter (BioSeb) and their tails were gently pulled until they released their grip. Grip strength was tested five times per mouse and maximum grip strength was recorded. For rotarod performance (Rotamex 5; Columbus Instruments), an initial speed of 4 r.p.m. was set, and this increased by 0.6 r.p.m. every 5 s until mice fell or was repeated (restarted) if a mouse held onto the axle for a complete revolution without falling. The longest time before falling of three independent attempts was recorded. Treadmill running capacity was determined using a motorized treadmill (Panlab/Harvard Apparatus). The treadmill was set at a 0° incline (flat) and initial speed of 10 cm per s, which was increased by 4 cm per s every 3 min until exhaustion. Mice were motivated to run by a shock grid at the rear of the belt, and exhaustion time was recorded when mice remained on the shock grid for >3 consecutive seconds.

In vivo glucose uptake

A subset of mice were injected i.p. with D-glucose (2 mg per kg) in saline containing 2.5% vol/vol [3H]-2-deoxyglucose (Perkin Elmer). Thirty min following injection, mice were euthanized via CO_2 inhalation and whole blood was collected via cardiac puncture into tubes containing EDTA. Tissues were rapidly dissected, weighed and snap frozen in liquid nitrogen. Glucose uptake into tissues was determined by scintillation as described previously[62] to determine total and unphosphorylated glucose, with the difference calculated as phosphorylated glucose.

Tissue collections and plasma separation

After 20 months of age, a subset of mice in each group were euthanized via CO_2 inhalation following a 4-h fast, whole blood was collected via cardiac puncture into tubes containing EDTA and tissues were rapidly dissected, weighed and either snap frozen in liquid nitrogen or fixed in formalin. Plasma was separated from whole blood by centrifuging at 6,000*g* for 5 min and then snap freezing in liquid nitrogen.

Plasma analyses

Plasma triglyceride, NEFA, ALT and fructosamine were determined using an automated enzymatic assay (COBAS Mira; Hoffmann-La Roche). Plasma insulin concentration was measured using an AlphaLISA immunoassay (Perkin Elmer). Plasma leptin and IGF1 were assayed using dedicated ELISA assay kits (AbcamK; leptin: ab199082; IGF1: ab100695). Protein carbonyls were determined using protein carbonyl assay kits (BPCK01; BioCell Corporation). Plasma cytokines were measured using a Bio-Plex Pro Mouse Cytokine 23-plex Luminex Assay (BioRad, Hercules). PI3Ki (alpelisib) concentration in plasma and liver homogenate were determined by the LC–MS/MS method described previously[25]. Briefly, liver samples were homogenized in two volumes of ice-cold ultrapure water (MilliQ) containing 0.1% formic acid using a TissueLyser II (Qiagen) set to an oscillation frequency of 30 Hz for 1 min. Plasma and liver homogenate samples (10 µL) were deproteinized by adding four volumes of ice-cold acetonitrile containing 0.25 µM ketoconazole as an internal standard, vortexed for 1 min and centrifuged at 20,000*g* for 5 min. The resulting supernatants were transferred to HPLC inserts and mixed 1:1 with ultrapure water (MilliQ) containing 0.1% formic acid. The ions were detected with electrospray ionization in positive multiple-reaction monitoring (MRM) mode. One MRM transition was utilized as a quantifier transition, and a second MRM transitions as a qualifier transition. Only one MRM transition was monitored for internal standard. The MRM m/z transitions monitored were 442 > 328 (quantifier) and 442 > 115 (qualifier) for alpelisib and 531 > 82 for ketoconazole.

Immunoblotting

Frozen tissue was weighed and homogenized in a tenfold volume of ice-cold RIPA lysis buffer supplemented with protease inhibitor cocktail (cOmplete mini EDTA-free protease inhibitor; Roche) and 2 mM PMSF before being mechanically homogenized (Tissue Lyser II; Qiagen). Total protein concentration was determined using a BCA-protein kit (Pierce BCA Protein Assay Kit, 23225; Thermo Fisher Scientific) with BSA as a standard. Protein (25–30 µg) was resolved using SDS–PAGE and transferred to either a PVDF or nitrocellulose membrane using a Transblot Turbo instrument (Bio-Rad Laboratories). Following 2 h of blocking in 2% fish gelatin in TBST, membranes were incubated overnight at 4 °C in primary antibodies. After the primary antibody was removed, membranes were washed in TBST and incubated for 1–2 h in HRP-conjugated secondary antibody diluted in TBST with 2% fish gelatine. Images were captured using a ChemiDoc MP Imaging System (Bio-Rad Laboratories) in the presence of chemiluminescent substrate (Clarity Western Substrate; Bio-Rad Laboratories), and bands were quantified using ImageLab 5.1 software (Bio-Rad Laboratories). Antibody details are provided in Extended Data Fig. 9a.

Gene expression

mRNA was isolated from frozen tissues using Quick-RNA MiniPrep Plus (Zymo Research) following the manufacturer's instructions. RNA concentration and quality were determined by spectrophotometry (Nanodrop One; Thermofisher Scientific) on mRNA from liver, subcutaneous adipose tissue and BAT. For microarray gene expression, mouse Clariom S Assay (Thermofisher Scientific) was used for BAT and female WAT samples, and Clariom D Assay was used for liver and male WAT. Only genes annotated as coding or multiple complex that were present in both Clariom S and D assays were used for analysis. Analysis and statistical comparisons were undertaken using the provided Transcriptome Analysis Console software (Thermofisher Scientific).

DNA damage

Hepatic nuclear and mitochondrial DNA lesions were quantified using the long amplicon quantitative PCR assay as previously described[63]. Briefly, DNA was extracted from liver using the AllPrep DNA/RNA/miRNA Universal kit (Qiagen) according to the manufacturer's instructions. DNA quality and purity were determined by spectrophotometry (Nanodrop One; Thermofisher Scientific; $A_{260}/A_{280} > 1.8$). DNA was quantified fluorometrically using PicoGreen (Thermofisher Scientific) according to the manufacturer's instructions. Fluorescence was measured with a 485-nm excitation filter and a 530-nm emission filter, and Lambda DNA/HindIII was used to construct a standard curve to determine the DNA concentration of unknown samples. PCR reactions were prepared by combining the following: 10 µl Platinum SuperFi II PCR Master Mix, 5 µl 3-ng µl^{-1} sample DNA (total 15 ng template), 0.5 µl of each 10 µM primer, and nuclease-free H_2O for a final volume of 25 µl. Primer sequences are outlined in Extended Data Fig. 9b. Each experiment included a no template control and a 50% control, which contained control DNA. These control samples ensured a lack of contamination in the reaction components and ensured that fragment amplification remained within the linear range. Each sample was vortexed briefly and subsequently centrifuged for 10 s before being amplified in a thermal cycler based on the conditions presented in Extended Data Fig. 9c.

To quantify products, PCR products were diluted one in ten in TE buffer, and 100 µl of the diluted PCR product was added to a white 96-well plate; 100 µl of the PicoGreen reagent was added to each well, and the plate was incubated in the dark at room temperature for 5 min. Fluorescence was measured with an excitation of 485 nm and emission of 530 nm. To control for differences in mitochondrial content between samples, large mitochondrial PCR products were normalized for copy number using fluorescence values of small mitochondrial PCR products as previously described[64].

Liver glycogen measurement

Liver glycogen content was determined by acid hydrolysis as previously described[65]. Briefly, duplicate sections of frozen liver were weighed and minced in either 2 M HCl (for determination of total glucose content with glycogen hydrolysis) or 2 M NaOH (for determination of free glucose). Minced samples were incubated at 95 °C for 1 h with constant shaking at 500 r.p.m. Samples were then cooled to room temperature and neutralized with either 2 M HCl or 2 M NaOH, following which they were vortexed and centrifuged at 22,000*g* for 20 min. Glucose concentration was determined from the resulting supernatants against a standard curve of D-glucose using hexokinase reagent (G3293; Sigma-Aldrich).

Adipocyte size quantification

Formalin-fixed adipose tissue was paraffin embedded, and H&E staining was performed on 5-µm sections cut using a microtome. Images of sections were captured using a Leica DMR upright light microscope with a ×10 objective lens. All images contained a 100-µm scale bar, which was used for calibration and subsequent adipocyte diameter determination using ImageJ. Adipocyte size frequency was determined, with adipocytes smaller than 20 µm or larger than 150 µm excluded from analysis[66].

Mitochondrial respiration

Mitochondrial respiration was determined from subcutaneous adipose tissue, visceral (epigonadal) adipose tissue and BAT from male mice. Adipose tissue was collected in ice-cold organ transplant buffer (Custodiol HTK) and stored on ice for 1–2 h before mitochondrial isolation. Mitochondria were isolated from adipose tissue by differential centrifugation. Tissue was minced in isolation buffer (250 mM sucrose, 5 mM HEPES, 1 mM EGTA, pH 7.4) and homogenized by six passes with a Teflon pestle in a tissue grinder. Tissue homogenates were centrifuged at 4 °C and 500*g* for 5 min to remove debris; then, the mitochondria-containing infranatant was decanted and centrifuged at 4 °C and 12,000*g* for 10 min to pellet mitochondria. Mitochondria were resuspended in isolation buffer with 1% wt/vol fraction V BSA. Mitochondrial respiration was measured in an Oroboros O2k oxygraph using a previously described protocol[26] with the addition of glycerol-3-phosphate. Briefly, oxygen consumption from isolated mitochondria was measured continuously at 37 °C with constant stirring at 750 r.p.m. Oxygen flux was determined following sequential addition of 2 mM malate + 5 mM pyruvate for electron flux through complex I, followed by 2.5 mM ADP to stimulate oxidative phosphorylation. Succinate (10 mM) was added to support respiration through complex II, followed by oligomycin (1 µM) to inhibit oxidative phosphorylation. We then added 5 mM glycerol-3-phosphate to support electron supply to complex III via mGPDH, followed by titrations of CCCP (0.5 µM) until maximum respiration was achieved. Complex III was inhibited using antimycin-a (2.5 µM) to account for any nonmitochondrial oxygen consumption, after which TMPD (0.5 mM) + ascorbate (2 mM) was added to measure maximal oxygen flux through complex IV. Auto-oxidation of TMPD + ascorbate was accounted for by inhibiting with 100 mM sodium azide, followed by reoxygenation of the respiration media. Data were collected and analyzed using Datlab 7 (Oroboros Instruments).

Ex vivo lipolysis and lipogenesis assays

Lipolysis and lipogenesis ex vivo were determined from duplicate explant (~20 mg) pieces of subcutaneous and epididymal fat from the same mouse, in Krebs Henseleit buffer containing 5 mM glucose and 3% wt/vol BSA. For lipolysis assays, explants were incubated with either isoproterenol (1 µM) or vehicle (PBS). For lipogenesis, 0.1 mCi/ml [3H]-D-glucose (Perkin Elmer) was added to the buffer with either 10 nM insulin or vehicle (PBS). Tissues were incubated for 2 h at 37 °C, after which the buffer was collected for measurement of NEFAs via enzymatic assay (COBAS Mira; Hoffmann-La Roche), and the tissue was collected, briefly rinsed in PBS and stored for later determination of lipogenesis. Lipogenesis was determined by homogenizing fat tissue in PBS using a rotor-stator homogenizer (Omni TH; Omni International), extracting lipids using the Folch et al. method[67], and then assaying for incorporation of [3H]-D-glucose into lipid by scintillation.

Statistics and reproducibility

Statistical analysis was performed using Prism v.8.0.0 (GraphPad Software Incorporated) or RStudio v.1.2.1335 statistical software (www.rstudio.com), with statistical significance determined as $P < 0.05$ and checked for normal distribution where appropriate. Specific statistical analysis used is indicated in the figure legends. Post hoc analysis was only undertaken for two-way ANOVA if a significant interaction effect was seen. No statistical method was used to predetermine sample size, but our sample sizes are similar to or greater than those reported in previous publications[7,8,11]. Treatment and vehicle condition were allocated by cage randomization based on body weight, ensuring before intervention that vehicle and treatment groups were matched for weight. Outliers were excluded based on a Grubbs test with α value set at a 0.05 significance level. Data distribution was assumed to be normal, but this was not formally tested. Investigators were not blinded to allocation during experiments and outcome assessment.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Source data for all figures and extended data are provided with this manuscript.

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Author contributions
C.P.H., B.S., S.C.B., C. MacRae, P.K., E.J.B., C. MacIndoe, T.T., B.G.M., S.S., M.A., J.K.J. and T.L.M. executed experiments. C.P.H., J.B., K.M.M., A.J.R.H., P.R.S. and T.L.M. contributed to the study design. T.L.M. wrote the manuscript, and all authors provided edits to the manuscript and assisted with interpretation of results.

Competing interests
The authors declare no competing interests.

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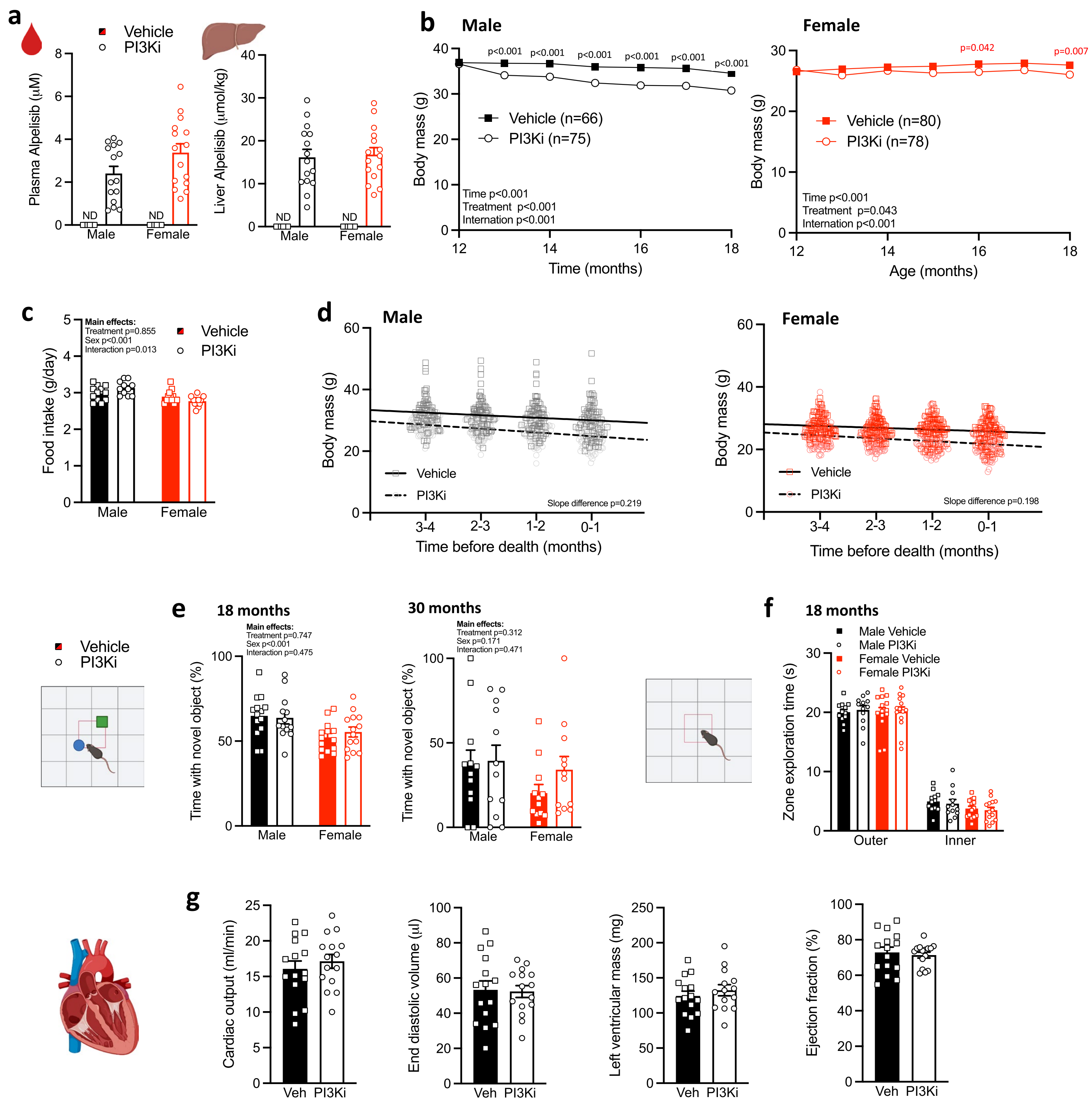
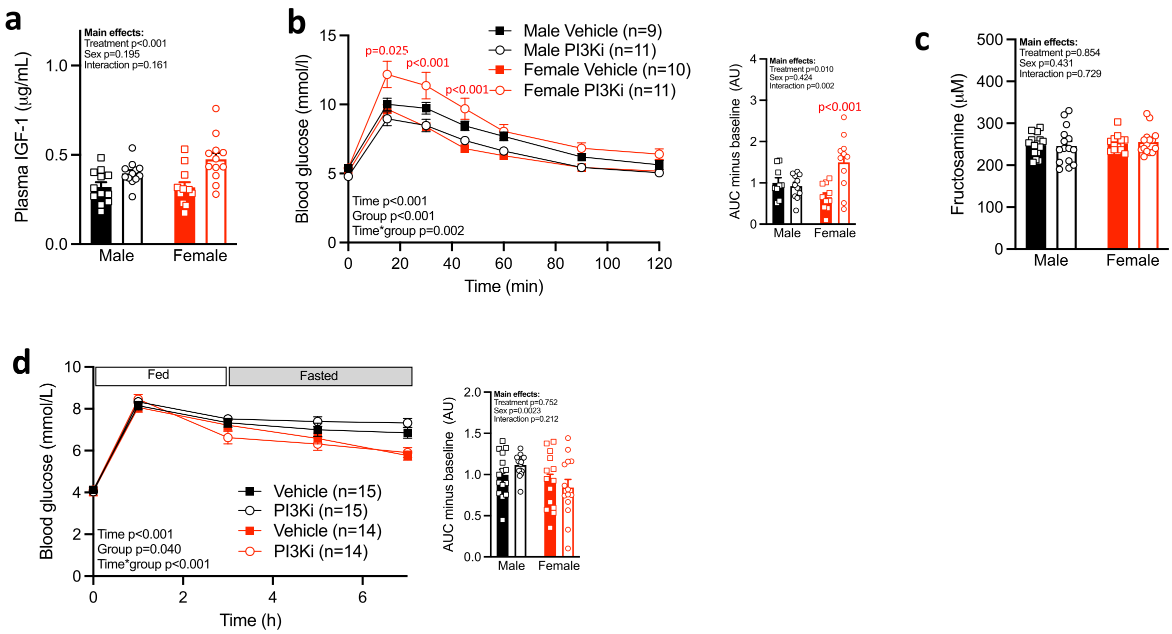


Extended Data Fig. 1 | Effect of PI3K inhibitor (PI3Ki) on body weight and cognitive and heart function in aging mice. a, Plasma and liver Alpelisib at 20 months of age. **b**, Body weights of mice that were alive for the entire first 6 months of treatment. **c**, Food intake at 15 months. **d**, Body weights during the last four months before death. **e**, Relative time spent with the novel object during a novel object test at 18 and 30 months, **f**, and time spent in the inner and outer zones during an open field test at 18 months. **g**, Echocardiograph measured cardiac output, end diastolic volume, left ventricular mass and ejection fraction in 15-month-old males. Results are mean ± SE with biological replicates (individual mice, n) shown as individual data points or stated within figure. Significance was determined by RM two-way ANOVA (a) two-way ANOVA (b, d, e), linear regression (c) or unpaired two-sided student t-test (f). P-values within figures for PI3Ki vs vehicle of the same sex.



Extended Data Fig. 2 | Effect of PI3K inhibitor (PI3Ki) on glucose homeostasis and insulin signalling with aging. a, Plasma IGF1 at 20 months. **b**, Glucose tolerance test at 30 months of age. **c**, Plasma Fructosamine at 20 months. **d**, Blood glucose response to 3 hours of ad libitum access to food (chow diet) following an overnight fast at 20 months. Results are mean ± SE with biological replicates (individual mice, n) shown as individual data points or stated within figure. Significance was determined by two-way ANOVA (a-d) or RM two-way ANOVA (b, d) with Sidak post-hoc analysis for within sex effects when a significant sex x treatment interaction was seen. P-values within figures for PI3Ki vs vehicle of the same sex.

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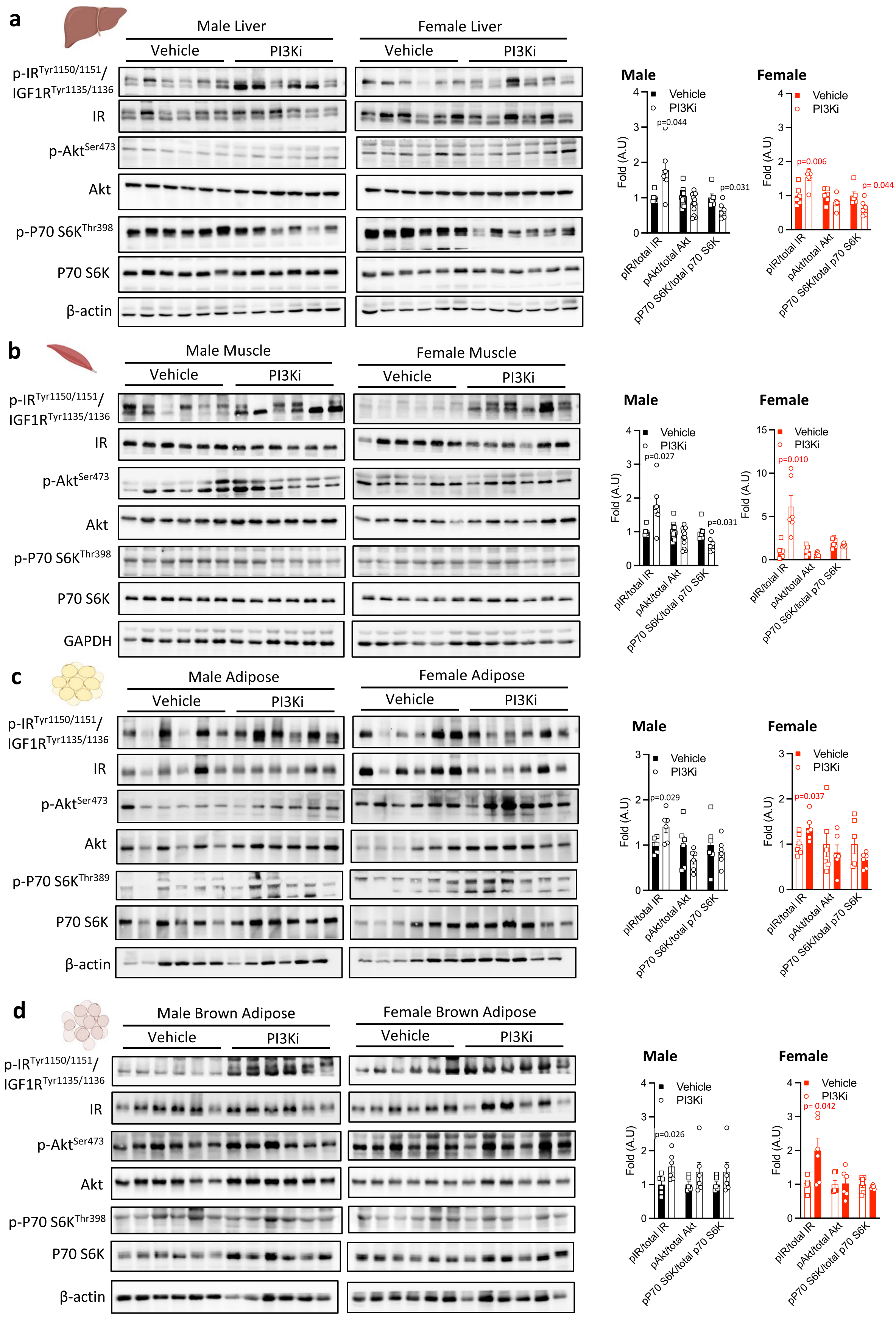

Extended Data Fig. 3 | See next page for caption.

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Extended Data Fig. 3 | Effect of PI3Ki on insulin signaling. Western blot images and quantification of insulin signalling intermediates in the, **a**, liver, **b**, muscle (gastrocnemius), **c**, white adipose tissue (WAT) and, **d**, brown adipose tissue (BAT) of 20-month-old mice. All samples blotted (n = 6 per group) are shown except for liver p-Akt^{Ser473} where blots are representative of n = 12 per group. Each well represents an independent mouse. Results are mean ± SE with biological replicates (individual mice, n) shown as individual data points. Significance was determined by an unpaired student two-sided t-test. P-values within figures for PI3Ki vs vehicle of the same sex.

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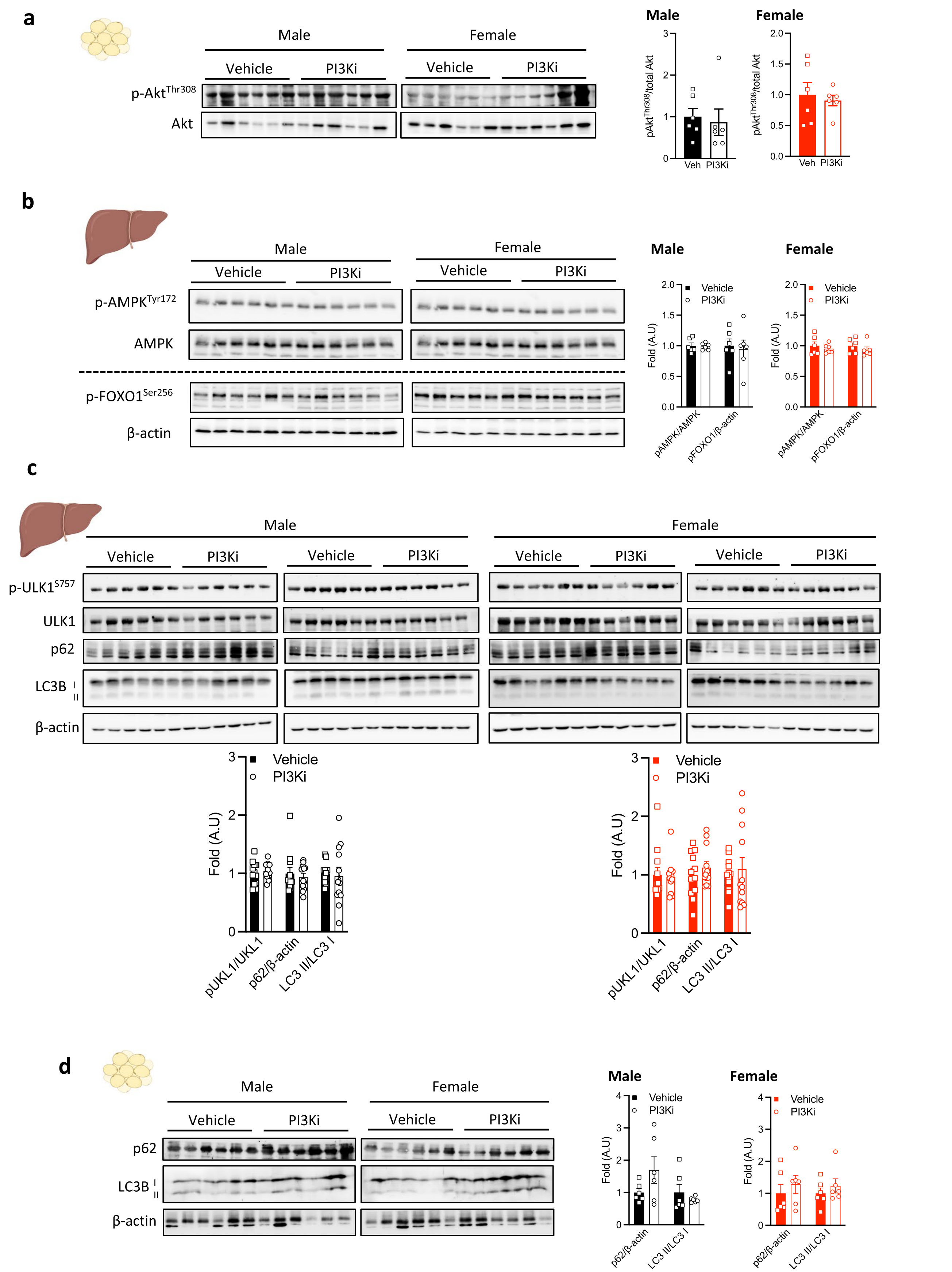
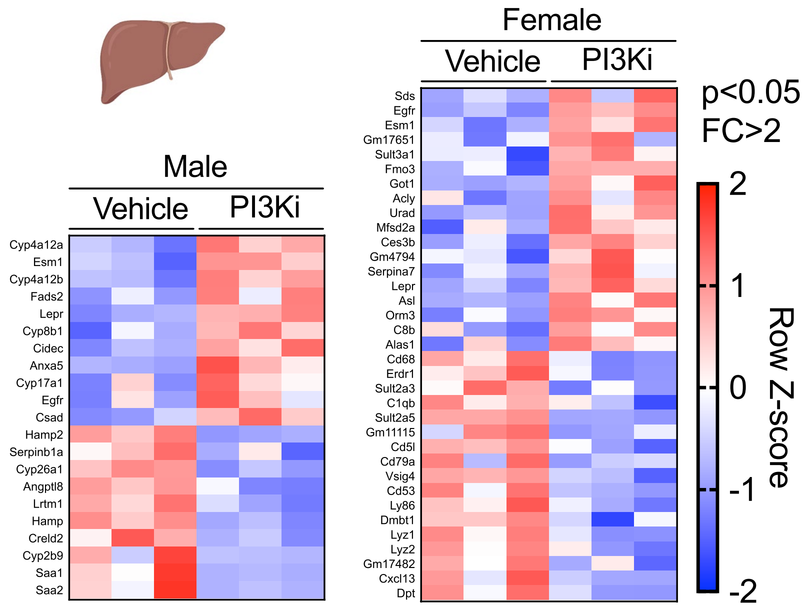


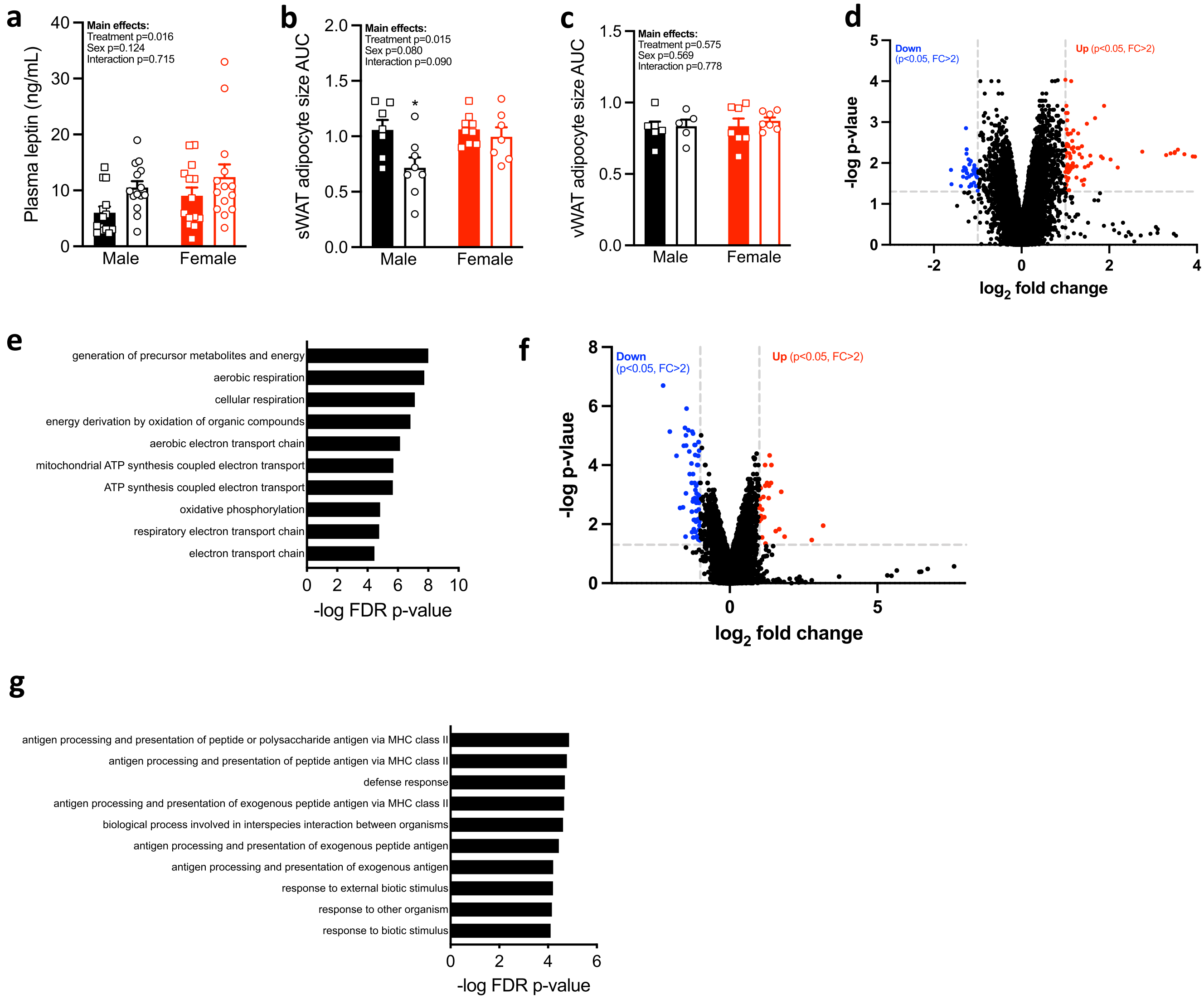
Extended Data Fig. 4 | Effect of PI3Ki on AktThr308, autophagy, AMPK and FOXO1 signalling. Western blot images and quantification of, **a**, white adipose tissue (WAT) Akt^{Thr308} phosphorylation (n = 6 per group), **b**, liver AMPK^{Tyr172} and FOXO1^{Ser256}, **c-d**, liver and WAT autophagy markers of 20-month-old mice. All samples blotted are shown. Results are mean ± SE with biological replicates (sample size, n) shown as individual data points. Significance was determined by an unpaired two-sided student t-test. P-values within figures for PI3Ki vs vehicle of the same sex.

 https://doi.org/10.1038/s43587-022-00349-y



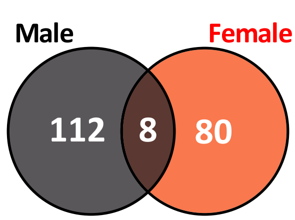
Extended Data Fig. 5 | Liver mRNA micro-array. Genes in the liver that expression was significantly altered by PI3Ki treatment at 20 months (by mRNA micro-array, n = 3 per group pooled from 2–3 independent mice).

 https://doi.org/10.1038/s43587-022-00349-y



h

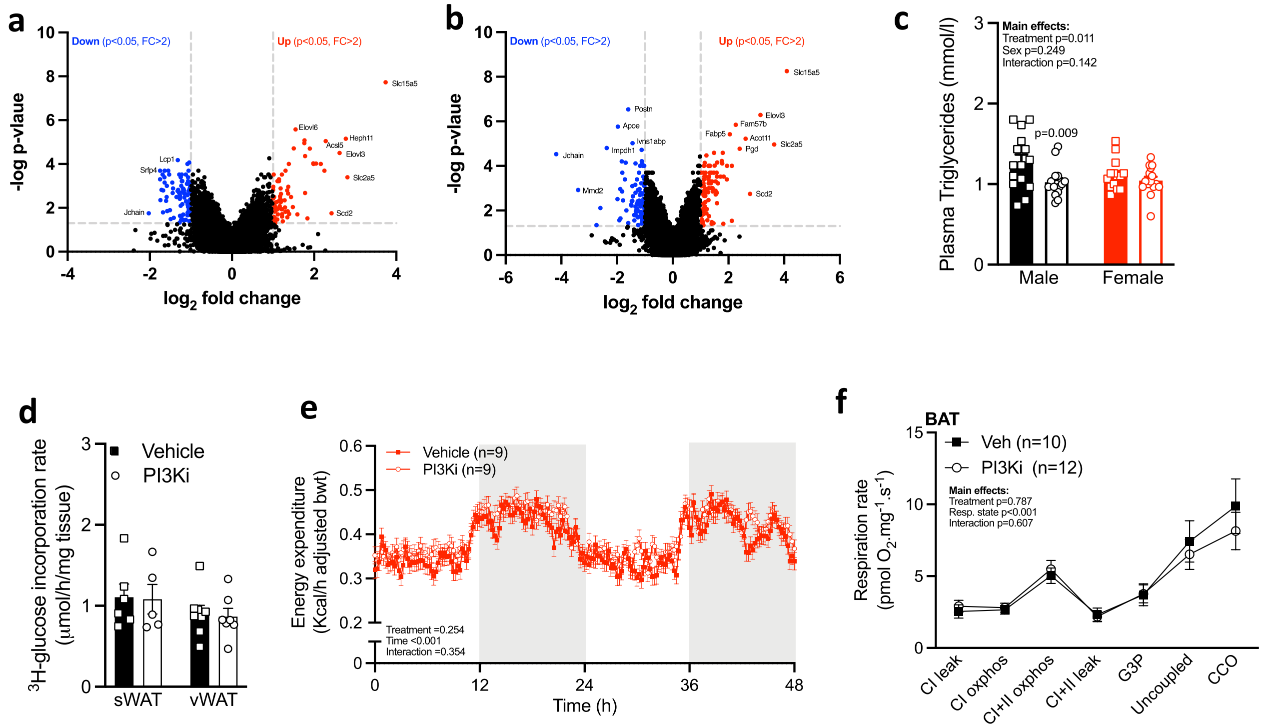
sWAT (p<0.05, FC>2)



Gene	Male FC	Female FC
Ifi27l2a	6.7	2.4
Arxes1	4.1	2.5
Hmgcs1	3.2	2
9330159F19Rik	2.7	2.1
Vnn1	-2	-2.3
Sfrp5	-2.2	-2.2
Cd209f	-2.5	-2.8
Cd209g	-2.6	-2.5

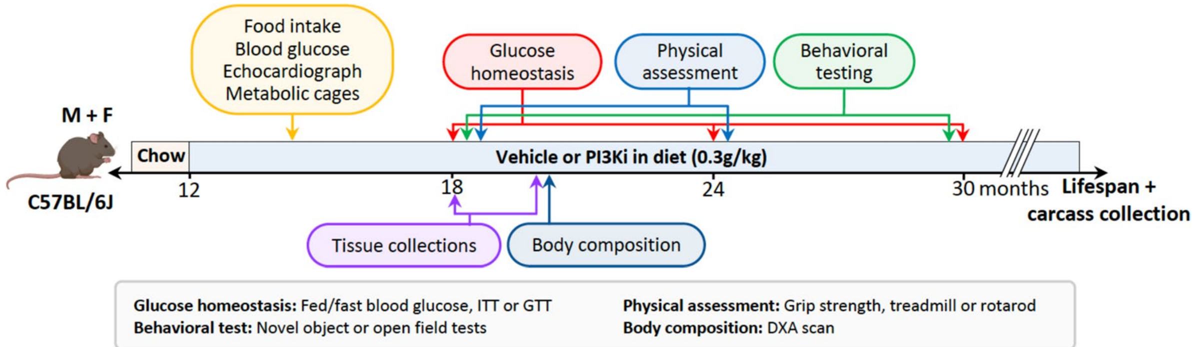
Extended Data Fig. 6 | Effect of PI3K inhibitor (PI3Ki) on white adipose tissue (WAT) during aging. a, Plasma leptin at 20 months. **b-c**, Subcutaneous (s)WAT and visceral (v)WAT area under adipocyte diameter curve (AUC; Fig. 3c). **d**, Male sWAT genes that expression was significantly altered by PI3Ki at 20 months of age (mRNA by micro-array, n = 4 per group pooled from 2–3 independent mice) and, **e**, associated top 10 gene ontology biological processes. **f**, Female sWAT genes that expression was significantly altered by PI3Ki at 20 months of age (mRNA by micro-array, n = 3 per group pooled from 2–3 independent mice) and, **g**, associated top 10 gene ontology biological processes. **i**, Significantly altered genes that overlap in male and female sWAT. Results are mean ± SE with biological replicates (sample size, n) shown as individual data points or stated within figure. Significance was determined by two-way ANOVA with sidak post-hoc analysis for within sex effects when a significant sex x treatment interaction was seen (a-c). P-values within figures for PI3Ki vs vehicle of the same sex.

https://doi.org/10.1038/s43587-022-00349-y



Extended Data Fig. 7 | Effect of PI3K inhibitor (PI3Ki) on brown adipose tissue (BAT) during aging. **a**, Male and, **b**, female BAT genes that expression was significantly altered by PI3Ki at 20 months of age (mRNA by micro-array, n = 3 per group pooled from 2–3 independent mice). **c**, Male and female plasma triglycerides, **d**, male isolated subcutaneous (s)WAT and visceral (v)WAT *de novo* lipogenesis at 20 months of age. **e**, Female Respiratory exchange ratio (RER) from 6 month-old-mice, and, **f**, mitochondrial oxygen consumption capacity of BAT from 20-month-old male mice. Results are mean ± SE with biological replicates (individual mice, n) shown as individual data points or stated within figure. Significance was determined by unpaired student t-test (c-d), mixed linear model (e) RM two-way ANOVA (f).

 https://doi.org/10.1038/s43587-022-00349-y



Extended Data Fig. 8 | Study design.

a

Target	Product code (supplier)	Lot number	Dilution used
$IR^{Y1150/1151}$ / $IGFR^{Y1135/1136}$	3024 (Cell Signalling Technology – CST)	11	1:1000
IR-β	3020 (CST)	6	1:1000
Akt^{S473}	3271 (CST)	14	1:1000
Akt^{T308}	9275 (CST)	27	1:1000
Pan-Akt	4691 (CST)	20	1:1000
$P70S6K^{T389}$	9205 (CST)	25	1:1000
P70S6K	2708 (CST)	7	1:1000
$AMPK^{T172}$	2531 (CST)	16	1:1000
AMPK	2532 (CST)	21	1:1000
$FOXO1^{S256}$	9461 (CST)	8	1:1000
$ULK1^{S757}$	6888 (CST)	5	1:1000
ULK1	8054 (CST)	7	1:1000
GAPDH	Ab9485 (Abcam)	GR3243682-3	1:5000
B-actin	A2228 (Sigma Aldrich)	067M4856V	1:5000
HRP-conjugated anti-Rabbit IgG	G21234 (ThermoFisher)	2273627	1:20000
HRP-conjugated anti-Mouse IgG	G21040 (ThermoFisher)	2043839	1:20000

b

Target	Forward primer	Reverse primer
8.7 kb fragment from the β-globin gene, accession number X14061	5′-TTG AGA CTG TGA TTG GCA ATG CCT-3′	5′-CCT TTA ATG CCC ATC CCG GAC T-3′
10 kb mitochondria fragment	5′-GCC AGC CTG ACC CAT AGC CAT ATT AT-3′	5′-GAG AGA TTT TAT GGG TGT ATT GCG G-3′
117 bp mitochondria fragment	5′-CCC AGC TAC TAC CAT CAT TCA AGT-3′	5′-GAT GGT TTG GGA GAT TGG TTG ATG-3′

c

Cycle step	β-globin	Large mito	Small mito
Initial denaturation	98°C for 30 s	98°C for 30 s	95°C for 5 min
Amplification:	20 cycles of:	18 cycles of:	20 cycles of:
Denaturation	98°C for 5 s	98°C for 5 s	95°C for 30 s
Annealing	60°C for 10 s	60°C for 10 s	60°C for 30 s
Extension	72°C for 4 min 20 s	72°C for 5 min	72°C for 45 s
Final extension	72°C for 5 s	72°C for 5 s	72°C for 7 min
Hold	4°C	4°C	4°C

Extended Data Fig. 9 | Extended method details. a) Details of antibodies used in immunoblotting, b) Gene targets and primer pairs for LA-qPCR, c) Thermal cycler conditions for amplification of nuclear and mitochondrial fragments.

nature portfolio

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Last updated by author(s): Oct 27, 2022

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☐	☒	A description of all covariates tested
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☐	☒	For null hypothesis testing, the test statistic (e.g. F, t, r) with confidence intervals, effect sizes, degrees of freedom and P value noted *Give P values as exact values whenever suitable.*
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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Lunar PIXImus II, Skyscan 1172 microCT, VisualSonics, MetaScreen, LabChart
Data analysis	Prism 8.0.0, RStudio v1.2.1335, VisualSonics, EthoVision XT 12 tracking software, Transcriptome Analysis Console (TAC) software, BioRender, ImageJ

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Reporting on sex and gender	N/A
Population characteristics	N/A
Recruitment	N/A
Ethics oversight	N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	n=76-87 per sex for lifespan study with in more that twice that of most lifespan study (n=20-30). For most other measures n>10 per group was used which is consistent with previous studies measuring similar phenomena. For example, based on previous data we preformed a sample size calculation to detect a 1 mmol/L difference in blood glucose. At an alpha-value of 0.05, 7 mice were required per group to detect a difference at a power of 80%, based on a standard deviation of 0.92 mmol/L.
Data exclusions	Outliers were excluded based on a Grubbs test with alpha-value set at 0.05.
Replication	Replication of the main finds of this manuscript were not possible due resourcing and time constrained- this was primarily a lifespan study in mice.
Randomization	To account for any housing and cohort variability, cages were randomly assigned to a treatment or control group. Within cage randomization was not possible as this study involved dietary intervention.
Blinding	Researchers were not blinded to the experimental due to ethical requirements for in vivo experiments. Due to the need for matching samples, aligning samples (western blot) and counterbalancing during analysis researchers were not blinded.

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Materials & experimental systems

n/a	Involved in the study	
☐	☒	Antibodies
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☒	☐	Palaeontology and archaeology
☐	☒	Animals and other organisms
☒	☐	Clinical data
☒	☐	Dual use research of concern

Methods

n/a	Involved in the study	
☒	☐	ChIP-seq
☒	☐	Flow cytometry
☒	☐	MRI-based neuroimaging

Antibodies

Antibodies used	All antibodies, including supplier and catalog number are listed in Extended Data Figure 8.

Validation	All antibodies were from commercial supplier which were validated by supplier and previous published reports. All were check for a band a correct molecular weight and, where applicable, total and phosphorylation antibody bands align. Please see Extended Data Figure 8 for catalog numbers to cross reference commercial supplier validation as required.

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals	Studies were undertaken in male and female C57bl6 mice bread in house. Age and given in methods and results section of manuscript
Wild animals	No wild animal were used in this research
Reporting on sex	Both male and female mice were used in this study. Results and analysis are reported individually for each sex.
Field-collected samples	No field collected samples were used in this research
Ethics oversight	All procedures were approved by the University of Auckland Animal Ethics Committee

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